Block 1 — Week 1

Foundations of Biomedical Informatics

Essentials of Biomedical Informatics — Self-Paced Module (Asynchronous)

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Overview

This is the first of four weekly modules in Block 1. It is self-paced: there are no live sessions. Plan for roughly 3–4 hours to read through the material, complete the walkthrough, and finish the exercise. Everything you need is in this guide plus two companion code files.

Companion Files

- Week1_Foundations_R.qmd — worked R/Quarto walkthrough + exercise
- Week1_Foundations_Python.ipynb — worked Python/Jupyter walkthrough + exercise

You only need to complete ONE language track (R or Python) to fulfill this week's deliverable, but you're welcome to try both — the course is designed to build comfort in both over time.

Learning Objectives

- Explain what biomedical informatics is and why standardized, observational health data matters
- Identify major types of health data (EHR, claims, patient-generated, genomic) and major national data initiatives
- Install and verify a working R/RStudio and Python environment
- Load a tabular dataset, compute descriptive statistics, and produce a simple visualization
- Render a reproducible report (Quarto or Jupyter) to HTML

Core Concepts

Biomedical informatics is the interdisciplinary field concerned with how information technology, data science, and computational methods can be applied to biomedical questions — from the molecular level (genomics) to individual patient care (clinical decision support) to population health (public health surveillance). It sits at the intersection of computer science, statistics, clinical medicine, and, increasingly, machine learning.

Health data comes in several major forms:

- Electronic health record (EHR) data — structured fields (diagnoses, medications, labs) and unstructured text (clinical notes) generated during routine care.
- Claims data — records generated by insurance billing; useful for utilization patterns but often less clinically detailed.
- Patient-generated data — wearables, patient-reported outcomes, and mobile health apps.
- Genomic and multi-omics data — sequencing and molecular data types, covered later in the course.

A key distinction runs through observational health research: in a randomized controlled trial (experimental), treatment assignment is controlled by the researcher, which isolates cause and effect. Most real-world health data is observational — patients and clinicians make decisions based on countless factors, and analysts must account for confounding and selection bias. These are recurring themes throughout the course.

Because health data is scattered across thousands of institutions in incompatible formats, several major initiatives now standardize and pool observational data at scale:

- All of Us Research Program (NIH) — a longitudinal cohort combining EHR, genomic, and survey data from over a million participants.
- National Clinical Cohort Collaborative (N3C) — a centralized, harmonized EHR repository built during the COVID-19 pandemic, now used broadly beyond COVID research.

- PCORnet — a national distributed research network connecting health systems for patient-centered outcomes research.

All three rely on common data models — most often the Observational Medical Outcomes Partnership (OMOP), which we dive into next week — to make data from different institutions jointly analyzable.

Finally, this course treats reproducibility as a practice, not an afterthought. Every exercise you submit should be something another person (or you, six months from now) could re-run and get the same result. That means version-controlled code, documented steps, and literate programming tools (Quarto, Jupyter) that interleave code, output, and narrative — exactly what you'll do in today's exercise.

Software Setup

All tools are free and open-source. Complete setup before starting the walkthrough.

1. Install R and RStudio

- Download and install R: https://cran.r-project.org
- Download and install the open source RStudio Desktop for your operating system: https://docs.posit.co/ide/user/#rstudio-ide-oss-downloads
- Open RStudio and run: `install.packages("tidyverse")`

2. Install Python

- Download Miniconda for your operating system: https://www.anaconda.com/download/success
- Verify installation in a terminal: `python3 --version`
- Install required packages: `pip install pandas numpy matplotlib jupyter nbconvert`

3. Install Quarto

- Download and install: https://quarto.org/docs/get-started/
- Verify installation in a terminal: `quarto check`

4. Verify your setup

- Open Week1_Foundations_R.qmd in RStudio and click Render — it should produce an HTML file with no errors.
- Open Week1_Foundations_Python.ipynb in Jupyter and run all cells — no errors should appear.

This Week's Deliverable

Complete the “Your Turn” section at the end of either companion file (R or Python):

1. Compute descriptive statistics for systolic_bp, split by smoker status.
2. Create one visualization comparing systolic_bp by smoker status.
3. Write 2–3 sentences interpreting the result, noting that this is synthetic data.
4. Render/export the file to HTML.

Submit both the source file (.qmd or .ipynb) and the rendered .html file via the Week 1 dropbox in Brightspace.

Self-Check Quiz

Use this to check your own understanding before submitting. Answer the quiz in Brightspace.

1. Which best describes “observational data” in health research?

- a) Data collected under a randomized controlled trial
- b) Data generated during routine care, without researcher-controlled treatment assignment
- c) Data that has been fully anonymized
- d) Data collected only from surveys

2. Which initiative links genomic, EHR, and survey data from over a million U.S. participants?

- a) N3C
- b) PCORnet
- c) All of Us Research Program
- d) MIMIC-III

3. Why do multi-institution networks like N3C rely on a common data model?

- a) To reduce data storage costs
- b) To make data from different institutions structurally comparable and jointly analyzable
- c) To anonymize patient identifiers automatically
- d) To comply with HIPAA automatically

4. Why did this week's exercise use a synthetic dataset instead of real patient data?

- a) Synthetic data is more statistically valid
- b) To practice workflows safely before working with real, standardized data structures
- c) Synthetic data is legally required for coursework
- d) R and Python cannot import real health data

5. What makes an analysis “reproducible”?

- a) It uses the most advanced algorithm available
- b) Another person can re-run the same code and steps and get the same result
- c) It has been peer reviewed
- d) It only uses point-and-click software

Suggested Readings

- Shortliffe, E. H., Cimino, J. J., & Chiang, M. F. (Eds.). (2021). Biomedical informatics : computer applications in health care and biomedicine (5th edition). Springer (chapter 1)
- What is Informatics? https://amia.org/about-amia/why-informatics/informatics-research-and-practice
- Observational Health Data Sciences and Informatics. (2021, January 11). The Book of OHDSI. https://ohdsi.github.io/TheBookOfOhdsi/ (chapters 1–2)
- All of Us Research Hub https://researchallofus.org/
- N3C https://ncats.nih.gov/research/research-activities/n3c/overview
- PCORnet https://pcornet.org/

Suggested Grading Rubric

A pass grade required 60/100 points

Criterion	Weight	What to look for
Environment setup	20%	File renders/executes without errors
Walkthrough completion	20%	All Part 1 cells/chunks run successfully
Task 1: grouped statistics	20%	Correct grouping logic and output
Task 2: visualization	20%	Appropriate chart type, labeled axes
Task 3: interpretation	20%	Shows understanding of synthetic vs. real data limits